SEMICON INDIA HACKATHON 2026

Team AIvengers — PS01 Idea Submission

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Problem Statement Addressed

AI-Based Restoration of Degraded Images for Semiconductor Inspection_KLA

Semiconductor Inspection & Quality Control Challenge:

- Microscopic inspection images are critical for chip defect detection and yield optimization.
- Low-dose Scanning Electron Microscopy (SEM) suffers from severe coupled image degradations.
- Conventional image filters smooth out critical micro-textures, hiding real wafer defects.

Three Official Degradation Categories (Joint Restoration Challenge):

- 1. Speckle Noise: Random pixel-level noise pushing intensity beyond standard signal bounds
- 2. Gaussian Noise: Reduces edge sharpness, blurring fine line/space grating structures
- 3. Resolution Reduction: Spatial downsampling (128x128 -> 256x256 2x Super-Resolution)

Idea Description: EDSR2x + L1 Loss

Lightweight, dataset-driven single-stage residual deep learning model

Model Architecture Specifications:

- Architecture: EDSR2x Residual CNN (Single-Stage)
- Residual Depth: 8 Residual Blocks (64 Feature Channels)
- Upsampling: 2x PixelShuffle Sub-Pixel Convolution
- Global Skip Connection: bicubic_2x(input) + learned_residual
- Parameters: 776,705 (~0.78M parameters)
- Training Loss: Pure L1 Reconstruction Loss
- Inference Latency: 2.95 ms / image (88.1 FPS on RTX 3050 GPU)

Why EDSR2x Was Selected (Controlled Research):

✓ Systematic evaluation across 6 distinct architecture candidates.
✓ Proved that complex window attention (SwinIR/HAT) amplified low-SNR noise.
✓ Proved that pre-denoising LR images lost high-frequency details.
✓ EDSR2x achieved the highest overall PSNR (27.42 dB) and SSIM (0.7357).

Proposed Solution: End-to-End Pipeline

Joint noise removal & 2x super-resolution system architecture

Our Solution: EDSR2x Neural Architecture

Single-stage deep residual network engineered for efficient 2x super-resolution

EDSR2x Model Specifications:

- 8 Residual Blocks (64 Feature Channels)
- 2x PixelShuffle Upsampling Layer
- Global Bicubic Skip Connection
- Parameters: 776,705 (~0.78M params)
- Training Loss: Pure L1 Reconstruction Loss
- Inference Latency: ~2.95 ms / image (RTX 3050 GPU)

```
NoisyLR (128x128)
   |
   ▼
[ Conv 3x3 (64 ch) ]
   |
   ▼
[ 8x ResBlocks + Conv ]
   |
   ▼
[ PixelShuffle 2x ] ──► [ Conv Tail ] ──► Learned Residual
                                               |
                                               + ◄── Bicubic 2x Skip
                                               |
                                               ▼
                                        Restored Output (256x256)
```

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EDSR2x Production Model

Innovation & Uniqueness

Dataset-driven engineering, degradation forensics, and lightweight optimization

1. Read-Only Degradation Forensics

Audited 3,200 paired images. Discovered Poisson-Gaussian shot noise ($\rho = +0.3955$) and direct 2x bicubic downsampling without heavy low-pass optical pre-blur.

2. Controlled Architecture Search

Tested EDSR2x, RCAN-Light, SwinIR-Light, HAT-Small, Wavelet-SR, and Multi-loss. Selected the simplest model with highest measured accuracy.

3. High-Frequency Bottleneck Identification

Identified that high-frequency noise overlap causes single-stage networks to over-smooth dense wafer contact holes, guiding optimal loss selection.

4. Avoiding Over-Processing

Demonstrated that multi-loss (Charbonnier + SSIM + Sobel) caused gradient conflicts (-2.22 dB PSNR loss). Pure L1 reconstruction loss achieved optimal performance.

5. Deployment-Oriented Optimization

0.78M parameters, 111.6 MB peak VRAM, and 2.95 ms GPU latency enable real-time inline semiconductor inspection integration.

Experimental Results & Visual Evidence

Quantitative benchmark and visual restoration comparison

Quantitative Performance Metrics

Rigorous evaluation across 320 official validation & 320 held-out test samples

Evaluation Split	Method	PSNR (dB)	SSIM	MAE	Edge Score	PSNR Win Rate
Validation (320 samples)	Bicubic 2x Baseline	22.59 dB	0.5166	0.0597	0.4727	—
Validation (320 samples)	**EDSR2x (Production)**	**27.42 dB**	**0.7357**	**0.0349**	**0.6639**	**320/320 (100%)**
Validation Net Gain	**EDSR2x vs Bicubic**	**+4.83 dB**	**+0.2191**	**-0.0248**	**+0.1912**	**100.0%**
Held-Out Test (320 samples	Bicubic 2x Baseline	22.98 dB	0.5243	0.0572	0.4781	—
Held-Out Test (320 sample	**EDSR2x (Production)**	**27.93 dB**	**0.7408**	**0.0310**	**0.6684**	**320/320 (100%)**
Held-Out Test Net Gain	**EDSR2x vs Bicubic**	**+4.95 dB**	**+0.2165**	**-0.0262**	**+0.1903**	**100.0%**

Technology Stack & Deployment Feasibility

Hardware throughput, standalone inference script, and memory audit

Model Efficiency & Hardware Latency

Engineered for real-time semiconductor line inspection integration

Parameter Count	Inference Latency	Batch Throughput	Peak GPU VRAM
776,705	2.95 ms	88.1 FPS	111.6 MB
~0.78M Parameters	per image (RTX 3050)	continuous processing	< 3% of 4GB VRAM

Designed for efficient, reproducible, high-throughput wafer defect inspection

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EDSR2x Production Model

GitHub Repository & Demo Video

Open-source codebase, reproducible packaging, and presentation video

Official Public GitHub Repository:

https://github.com/TiyasDas-81/Semicon_Hackathon_26.git

Official Presentation Demo Video:

File Name: EDSR2x_demo.mp4 (Included in fresh_training/final_submission/)

Video Attributes: 83 seconds duration | 1920x1080 Full HD | 30 FPS H.264 MP4

Official Submission Entry Point:

python run.py <input-dir> <output-dir>

Processes all .npy inputs and generates corresponding restored .npy outputs.

References & Academic Citations

Authoritative literature and official competition problem specifications

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6. i4C & KLA SemiCon India Hackathon 2026. Problem Statement PS01: AI-Based Restoration of Degraded Images for Semiconductor Inspection.